OMNIA VANITAS STUDIOS

LIKENESS : THE GAME

V002 EXPANSION MODULES

Image-Blaster - Nano Gaussian Splatting - Higgsfield Claude Skills - Unreal Engine Guide

Purpose: add the requested workflows as controlled optional enhancers to the existing LIKENESS pipeline without destabilizing the main Unreal game, Three.js companion, 4-step character-reference workflow, or OVS production-bible standard.

Production law: Expansion modules prototype, enrich, test, and market. They do not replace the core authored story, locked identities, game architecture, or shared visual thread.

Expansion Source Matrix - LIKENESS : THE GAME V002

These workflows are added as optional expansion modules only. They enhance the existing OVS production pipeline without replacing the Unreal vertical-slice foundation, the 4-step reference system, or the Three.js/Fable companion architecture.

Module	Source	Use In LIKENESS	Status	Gate
Image-Blaster	https://github.com/bthoreau88/image-blaster	Rapid environmental prototyping, splat + mesh + ambient SFX experiments from a single image	Optional	Never replaces authored hero level design. Must pass scale, collision, mood, and optimization QC.
Nano Gaussian Splatting	https://github.com/TimChen1383/NanoGaussianSplatting	Unreal splat rendering test layer for captured/generated static environments	Optional / Technical Pilot	Use only for static environment visuals. Pair with invisible collision mesh. Slice large splats. Validate FPS and ghosting.
Higgsfield Claude Skills	https://github.com/AKCodez/higgsfield-claude-skills	Campaign-video, lookbook, mood, reference-video, social-hook expansion workflow	Optional / Marketing + Reference	Keep OVS visual DNA. Do not allow UGC defaults to overwrite LIKENESS tone. Confirmation before generation.
Unreal Engine Guide	https://github.com/mikeroyal/Unreal-Engine-Guide	Learning reference and checklist companion for UE concepts, tools, rendering, PCG, VFX, C++, Blueprint, platform notes	Reference Only	Do not clone as source code dependency. Use as study and checklist index.

Source Notes

- Image-Blaster describes an image-to-world Claude skillset that creates GLB/OBJ dynamic object models, SPZ Gaussian splats for static environments, and MP3 ambient/SFX outputs from an input image. It uses World Labs Marble, FAL/Hunyuan 3D, image cleanup/editing models, and ElevenLabs SFX.
- Nano Gaussian Splatting supports UE5.6 and UE5.7, imports PLY files into Gaussian Splat Assets, and uses Nanite-style LOD clusters, screen-space error selection, splat compaction, and GPU sorting techniques for large splat rendering.
- Higgsfield Claude Skills provides 19 Claude Code slash-command skills with Playwright automation, including cinematic, 3D CGI, music video, social hook, brand story, and fashion lookbook prompt workflows.
- Unreal Engine Guide is an external reference index for Unreal Engine tools, learning resources, Blueprint, Lumen, Nanite, PCG, Niagara, MetaHuman, C++, Python, and related developer topics.

12 - OVS Expansion Workflow Modules

Purpose

These modules are not the base game. They are controlled accelerators for the LIKENESS ecosystem: Unreal vertical slice, Three.js interactive web companion, OVS marketing rollout, and reference/asset production. The existing core workflow remains the source of truth.

Expansion Law

1. Core story and gameplay remain authored first.
2. AI-generated worlds are prototypes until art-directed and optimized.
3. Gaussian splats are visual surfaces, not gameplay truth.
4. Campaign-video skills are marketing/reference tools, not tone directors.
5. External guides support learning and checklist coverage; they do not become project authority.
6. Every expansion asset must enter staging, pass QC, then move to approved.
7. Every agent must report what changed, what it did not touch, and what needs human review.

The Four Added Expansion Tracks

Track A - Image-Blaster Rapid World Prototyping

Best use cases:

- Quick Room 14 alternate layout experiments.
- Dellwood motel exterior mood studies.
- Lobby, hallway, bathroom, stairwell, parking lot, and dream-space previsualization.
- Ambient room tone and object-specific SFX placeholders.
- Three.js companion prototypes where speed matters more than final fidelity.

Do not use for:

- Final hero geometry without cleanup.
- Character identity work.
- Gameplay collision without validation.
- Anything that must preserve exact floor-plan continuity.

Pipeline:

1. Place one clean OVS-approved concept image into `/expansions/image_blaster/input/`.
2. Ask the agent to blast it with confirmation at each step.
3. Collect generated `.spz`, `.glb`, `.obj`, `.mp3`, preview screenshots, and metadata.
4. Move outputs into `/expansions/_staging/image_blaster/<date>_<scene>/`.
5. Run QC: scale, collision, lighting, mood, file size, polygon count, texture naming, legal/licensing note.
6. Convert approved static environment splat into either a Three.js Spark.js test or an Unreal NanoGS test.
7. Rebuild final hero level manually when the scene becomes canonical.

Track B - Nano Gaussian Splatting Unreal Test Layer

Best use cases:

- Static haunted environment capture layers.
- Real-world scan experiments.
- Browser-to-Unreal parity tests.
- Optional photogrammetry mood backgrounds.

Do not use for:

- Dynamic destructible gameplay.
- Interactive props requiring precise collision.
- Character rendering.
- Lighting-critical scenes that require full Lumen response.

Pipeline:

1. Install the plugin in the Unreal project's `Plugins/` folder.
2. Use UE5.6/UE5.7-compatible builds only.
3. Import `.ply` splat files and generate Gaussian Splat Assets.
4. Drag assets into a test map, never directly into the production Room 14 map first.
5. Pair splat visual with invisible collision mesh or authored blocker geometry.
6. Use tiled/sliced splats when scenes are large.
7. Test console commands and performance settings.
8. Capture before/after screenshots and FPS notes.

Track C - Higgsfield Claude Skills Campaign + Reference Expansion

Best use cases:

- OVS social hooks.
- LIKENESS teaser reels.
- Fashion/lookbook experiments for OVS wardrobe language.
- Cinematic motion-reference prompts.
- Marketing variations for the film/game/site shared universe.

Do not use for:

- Replacing the locked character identity sheets.
- Turning LIKENESS into generic UGC style.
- Any content that violates the project's restraint/no-gore law.
- Unapproved automated posting or generation.

Pipeline:

1. Install skills only in a separate marketing/reference workspace.
2. Add an OVS-specific `CLAUDE.md` override.

3. Use only approved style commands: `/01-cinematic`, `/02-3d-cgi`, `/10-music-video`, `/11-social-hook`, `/12-brand-story`, `/13-fashion-lookbook`.
4. Rewrite every prompt through the LIKENESS visual DNA before generating.
5. Require screenshot confirmation before clicking Generate.
6. Store outputs in `/expansions/_staging/higgsfield/<campaign>/`.
7. Promote only prompt text, motion ideas, and approved still/video outputs into `/assets/`.

Track D - Unreal Engine Guide Learning + Checklist Expansion

Best use cases:

- Building a study roadmap.
- Checking if a feature belongs in Blueprint, C++, Niagara, PCG, Lumen, Nanite, MetaHuman, or Python.
- Creating a weekly learning board for Project and Portfolio IV alignment.
- Expanding the production bible with missing technical topics.

Do not use for:

- Blindly copying code.
- Treating the guide as official Epic documentation.
- Replacing project-specific architecture decisions.

Pipeline:

1. Use it as a reference index.
2. Convert relevant topics into LIKENESS checklists.
3. Link every chosen concept to a concrete project need.
4. Validate implementation details against official docs or current engine behavior.
5. Add lessons learned to `/build_notes/` and `TASK_BOARD.md`.

Expansion Promotion Gate

An output moves from expansion to production only if it passes:

- Story fit.
- Visual DNA fit.
- Technical viability.
- Performance sanity.
- File organization compliance.
- Naming convention compliance.
- License/source note.
- Human approval.

Folder Routing

- Experimental outputs: `/expansions/_staging/`
- Approved expansion assets: `/expansions/_approved/`

- Rejected or obsolete tests: `/expansions/_archive/`
- Canonical production assets: `/assets/`
- Registries: `/data/`
- Agent instructions: `/agents/`

13 - Gaussian Splat and Image-to-World Optional Module

Role in LIKENESS

Gaussian splats are an optional visual layer for location memory, haunted-space realism, and rapid environment prototyping. They are not the default gameplay substrate. In LIKENESS, they work best as static mood environments paired with clean collision meshes, authored interaction zones, and data-driven narrative triggers.

Accepted Inputs

- `.spz` from image-to-world tools.
- `.ply` splats from scans or generators.
- `.glb` or `.obj` dynamic object meshes.
- `.mp3` ambience or SFX placeholders.
- Preview screenshots.
- Metadata JSON or markdown notes.

Unreal Path

1. Keep the canonical Unreal game in authored UE maps.
2. Create separate test maps named `LVL_TEST_SPLAT_<SCENE>`.
3. Import PLY splats through Nano Gaussian Splatting only after the plugin is isolated in a test branch.
4. Use invisible collision meshes or authored blocking volumes.
5. Never rely on splat pixels for collisions, navmesh, or interaction traces.
6. Test with target camera paths: third-person exploration, over-the-shoulder, static witness camera, and inspection close-up.
7. Record FPS, VRAM behavior, visible ghosting, TSR artifacts, and LOD/culling behavior.

Three.js / Web Companion Path

1. Keep the Three.js companion lightweight and route-based.
2. Use splats for atmosphere, interactive memory dioramas, or web exploration scenes.
3. Use mesh colliders for all interactions.
4. Store splat manifest metadata separately from React state.
5. Keep fallback image/card mode for phones and low-memory devices.

Optimization Rules

- Slice large splats into tiles or scene zones.
- Delete nearly transparent splats if they cause ghosting.
- Use authored occluder/collision meshes.
- Keep dynamic props as meshes, not splats.
- Build lowest viable test first.
- Never ship a splat level without fallback screenshots or baked panorama alternative.

LIKENESS Scene Candidates

Scene	Best Use	Notes
Dellwood exterior	Web mood diorama	Good for rainy motel memory space.
Room 14 alternate memory	Unreal test only	Must not replace canonical room layout.
Bathroom mirror	Experimental	Splat can hold environment, mirror logic remains authored.
Hallway five/six doors	Strong candidate	Static haunted corridor fits splat mood.
Parking lot at 3:14 AM	Strong candidate	Good for atmospheric web exploration.
Family-photo wall	Weak candidate	Better authored by hand for exact props.

Agent Acceptance Prompt

When using image-to-world or splat tools, the agent must answer:

1. What was the input image?
2. What files were produced?
3. Which files are splats, meshes, sound, or metadata?
4. What belongs in Unreal, Three.js, Blender, or archive?
5. What is missing for collision?
6. What is missing for lighting?
7. What is the file-size and performance risk?
8. What should be manually rebuilt before production?

14 - Higgsfield / Seedance Campaign and Reference Expansion

Role in LIKENESS

This module supports reference video, campaign motion, social hooks, lookbook tests, and pitch materials. It does not define the game, story, or character identity. Every output must be rewritten through OVS visual DNA before it is allowed into the production ecosystem.

Approved OVS Uses

- Social teaser reels for LIKENESS.
- Wardrobe-motion tests for Thoreau, Drya, doubles, and OVS fashion drops.
- Music-video fragments tied to score/sound design.
- Web landing page hero loops.
- Proof-of-tone clips for investors, collaborators, or class presentations.
- Motion prompts for future animation vendors.

Restricted Uses

- No generic UGC influencer voice for main LIKENESS scenes.
- No face drift from locked identity sheets.
- No automatic generation without confirmation.
- No unapproved sexualized, exploitative, gore, or violence escalation.
- No platform-native style overriding OVS palette, typography, or restraint.

OVS Higgsfield CLAUDE.md Override

```
# OVS LIKENESS Higgsfield Rules Default project: LIKENESS : THE GAME / OVS universe. Default
tone: psychological romantic horror, cinematic restraint, no gore escalation. Default image
language: ARRI Alexa 35 / Mini LF, Cooke or Panavision anamorphic, warm skin, cyan-green
shadows, tungsten/teal conflict, 35mm grain, shallow but readable depth of field. Default UI
text: OVS SHORT FILM / LIKENESS / THE GAME / OMNIA VANITAS STUDIOS. Default character rule:
locked identities always come from approved reference sheets, never invented. Default
confirmation rule: screenshot before generation; ask for approval before clicking Generate.
Default storage: save outputs to /expansions/_staging/higgsfield/<campaign_name>/. Allowed
commands: /01-cinematic, /02-3d-cgi, /10-music-video, /11-social-hook, /12-brand-story,
/13-fashion-lookbook. Disallowed tone: generic UGC ad, bright influencer comedy, plastic CGI,
thirst-trap campaign, gore spectacle.
```

Campaign Prompt Template

```
Use the OVS LIKENESS visual DNA. Build a [duration] [format] motion prompt for
[scene/campaign]. Character identity source: [reference sheet name]. Scene: [Room 14 / hallway
/ exterior / web companion / wardrobe test]. Emotional beat: [what the viewer should feel].
Camera: [lens, movement, framing]. Lighting: [tungsten, teal, practical, haze]. Motion: [subtle
action only]. Text overlay: [OVS hierarchy]. Negative: no identity drift, no plastic skin, no
```

```
generic UGC tone, no gore escalation, no random wardrobe changes. Ask for confirmation before
generation.
```

Output Review Checklist

- Does the face stay locked?
- Does the color grade feel like LIKENESS?
- Does motion feel like a film/game artifact, not an ad template?
- Does the prompt preserve story restraint?
- Does the output fit Unreal, Three.js, IG, or pitch use?
- Is the file stored in staging with a note?
- Is a still frame usable as a production reference?

15 - Unreal Engine Guide Reference Expansion

Role

The Unreal Engine Guide repository is treated as a learning/reference index. It helps the LIKENESS team identify relevant systems, but it does not override Epic documentation, engine behavior, or project-specific architecture.

LIKENESS Mapping

Guide Topic	LIKENESS Use
Blueprint Visual Scripting	Interaction triggers, quick prototypes, room-state logic.
C++	Stable gameplay systems, save system, interaction interfaces, data parsing.
Lumen	Mood lighting, tungsten/teal conflict, indirect light tests.
Nanite	High-fidelity props and geometry where appropriate.
Niagara	Dust, rain, breath, memory distortion, subtle mirror particles.
PCG	Motel exterior clutter, hallway variation, environmental dressing.
MetaHuman	Possible character prototype path only after identity lock is stable.
Python	Batch import, registry validation, editor automation.
Photogrammetry / LiDAR	Optional location capture or scan-based environment reference.
C++ / Visual Studio	Build reliability and source-control discipline.

Weekly Study Sprint Template

1. Pick one feature tied to a production need.
2. Create one small test map or sandbox asset.
3. Document what worked and failed.
4. Add an implementation decision to the production bible.
5. Commit the sandbox separately from production content.

Example Sprints

- Sprint 01: Interaction interface and inspectable object pickup.
- Sprint 02: Lumen motel lighting study.
- Sprint 03: Niagara dust/rain/breath layer.
- Sprint 04: PCG hallway clutter generator.
- Sprint 05: Python asset registry validator.
- Sprint 06: Save/load room-state prototype.
- Sprint 07: Camera witness mode and inspection camera.
- Sprint 08: Performance pass and stat captures.

Rule

A learning item becomes part of LIKENESS only when it solves a named production problem.

Agent Prompt - Expansion Modules Only

Use this prompt when handing the V002 expansion package to Claude Code, Codex, or another agent.

```
You are working inside LIKENESS : THE GAME / OVS production package V002. Your task is to add
or test expansion workflows only. Do not rewrite the core pipeline unless a human explicitly
asks. Read these files first: - README_START_HERE.md - agents/AGENTS.md -
docs/12_Expansion_Source_Matrix.md - docs/12_Expansion_Workflow_Modules.md -
docs/13_Gaussian_Splat_Image_To_World_Module.md -
docs/14_Higgsfield_Campaign_Reference_Module.md -
docs/15_Unreal_Engine_Guide_Reference_Module.md - data/expansion_registry.csv Expansion rules:
1. Treat Image-Blaster, Nano Gaussian Splatting, Higgsfield Skills, and Unreal Engine Guide as
optional enhancers. 2. Keep core Unreal game, Three.js companion, 4-step character reference
system, and OVS style bible intact. 3. Put all tests into /expansions/_staging/ first. 4. Do
not promote files into /assets/ until QC passes. 5. Do not add API keys to git. 6. Do not auto-
generate or click external generation buttons without confirmation. 7. Do not use splats as
collision or gameplay truth. 8. Do not let generic UGC/social styles override LIKENESS visual
DNA. 9. After every change, update SESSION_RESUME.md and TASK_BOARD.md. 10. Report exact files
changed, exact commands run, exact assets staged, exact risks found, and exact next step.
First action: Create a branch named expansion-modules-test, verify folder structure, read
expansion_registry.csv, and produce a short implementation plan for one safe test: either
Image-Blaster staging, NanoGS plugin test map, Higgsfield prompt-only campaign test, or Unreal
learning checklist update. Do not install anything until the human approves the chosen test.
```

Implementation Checklist

[ ] Keep all expansion work in a branch or staging folder until approved.

[ ] Never commit API keys or private generated outputs with unclear licensing.

[ ] Always capture screenshots and notes from splat/render tests.

[ ] Promote assets only after OVS QC and registry update.

[ ] Use Higgsfield/Seedance only as prompt/reference/campaign expansion with confirmation.

[ ] Use Unreal Engine Guide as a learning reference, not a dependency.

OVS Footer Note

This V002 expansion pack is additive. The V001 Production Bible remains valid. Use this document when you want faster prototype worlds, splat rendering tests, campaign motion prompts, and structured Unreal learning support.